MUZNA SIDDIQUI

(22F-BSAI-37)

DEPARTMENT OF ARTIFICIAL INTELLIGENCE

DEEP LEARNING ASSIGNMENT REPORT : "Building an Intelligent Clinical Early Warning System"

SIR HAMZA FAROOQUI

Section 1: The Clinical Problem

1.1 Why Patient Deterioration Prediction is Uniquely Difficult

Predicting patient deterioration in a hospital setting is fundamentally more complex than a standard classification task for several reasons. First, patient physiology evolves continuously and non-linearly. A deteriorating patient does not flip a binary switch from "stable" to "sick"; instead, their condition drifts across dozens of interacting biomarkers over hours or days. A model that only sees a single snapshot of vital signs cannot detect these temporal trajectories. Second, the class imbalance is extreme and clinically motivated. In the PhysioNet Sepsis Challenge dataset, only approximately 8% of patient-hour records correspond to a positive (sepsis) label. Standard classification algorithms, optimised for accuracy, will learn to simply predict the majority class. Third, clinical data is notoriously incomplete. Lab values such as Lactate, pH, and Creatinine are measured only when ordered by a clinician, leaving 30 to 40 percent of feature entries as missing values. Any model must be robust to this pattern of missingness, which is not random but informatively absent.

1.2 Why Tabular Vital Signs Alone Are Insufficient

Structured vital signs provide a quantitative snapshot of a patient's physiological state: heart rate, blood pressure, temperature, oxygen saturation. These numbers, however, cannot capture the clinical reasoning of a physician at the bedside. A patient whose lactate is rising but whose chart note reads "fluid resuscitation ongoing, antibiotics administered, improving" is in a fundamentally different risk category than a patient with identical numbers and no documented treatment. Unstructured clinical notes encode a physician's differential diagnosis, the trajectory of the patient's condition as perceived by the care team, the presence of co-morbidities, and the urgency of the language used. Phrases such as "concerning for sepsis," "broad-spectrum antibiotics initiated," or "rapid response called" carry clinical signal that no numeric feature can represent. The integration of free-text notes is therefore not optional but essential for a complete early warning system.

1.3 Why Recall Matters More Than Accuracy

In clinical risk prediction, the cost of error is asymmetric. A false positive means the model flags a patient as high-risk when they are actually stable. The consequence is an unnecessary clinical review: a physician spends a few minutes re-examining the patient. A false negative means the model classifies a deteriorating sepsis patient as stable. The consequence is that no intervention is triggered. Given that untreated sepsis progresses to septic shock within hours and that mortality risk doubles with every hour of delayed treatment, a missed case can cost a life.

Consider the following scenario: Patient A, a 67-year-old admitted for a urinary tract infection, begins showing a rising heart rate and declining MAP at 2 AM. The model, optimised for accuracy, assigns a 0.48 probability of sepsis and does not trigger an alert (threshold 0.5). By 6 AM, the patient is in septic shock requiring ICU transfer. In this scenario, a model with 90% accuracy but 60% recall is clinically inferior to a model with 85% accuracy and 92% recall. This is why our implementation uses a decision threshold of 0.45 rather than the conventional 0.5: we deliberately trade some precision for higher recall, accepting that more false alarms are preferable to missed cases in a life-critical system.

Section 2: Baseline Design Decisions

2.1 The Vanishing Gradient Problem and Mitigation

The vanishing gradient problem arises during backpropagation through deep networks. As gradients are multiplied layer by layer via the chain rule, values smaller than one are multiplied repeatedly and become exponentially small by the time they reach earlier layers. As a result, the weights of early layers receive negligible updates and the network fails to learn long-range feature interactions. The severity of the problem depends directly on the choice of activation function.

In our DNN implementation, we use the Rectified Linear Unit (ReLU) activation, which outputs zero for negative inputs and the identity for positive inputs. The gradient of ReLU is either zero or one, preventing multiplicative shrinkage for active neurons. This is in contrast to sigmoid or tanh activations, whose derivatives are always less than one and contribute to vanishing gradients in deep networks.

Batch Normalization further addresses this problem by standardising the activations of each layer to have zero mean and unit variance. This prevents the activation distribution from drifting during training, a phenomenon known as internal covariate shift. Normalised activations ensure that gradients flowing through the network remain in a healthy range, allowing the use of larger learning rates and accelerating convergence. In the clinical context, Batch Normalization is especially important because features such as Creatinine and Lactate have very different natural scales and clinical significance, and even after StandardScaler preprocessing, their post-ReLU distributions can diverge.

2.2 Optimizer Comparison: SGD vs. Adam

Two optimizers were compared in the Generation 1 experiments. Stochastic Gradient Descent with momentum updates parameters in the direction of the gradient, accumulating velocity to navigate ravines in the loss surface. It is robust and theoretically well-understood but requires careful learning rate tuning and converges slowly, particularly on sparse or noisy gradients characteristic of clinical data.

Adam (Adaptive Moment Estimation) maintains per-parameter learning rates by tracking both the first moment (mean of gradients) and the second moment (uncentered variance). This adaptive scaling allows Adam to make larger updates for infrequent features and smaller updates for frequent ones, making it particularly well-suited to datasets with missing values and class imbalance. In our experiments, Adam converged significantly faster and achieved higher recall, making it the preferred optimizer for the baseline model.

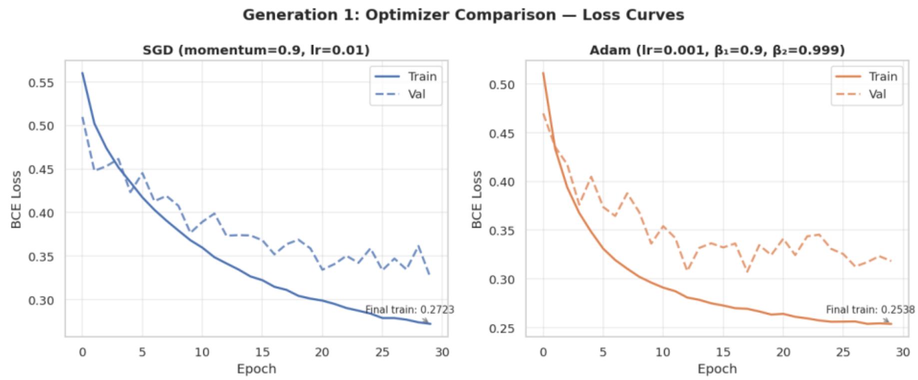

Figure 1: Loss curves comparison between SGD and Adam optimizers over 30 training epochs. Adam achieves faster convergence and lower final validation loss.

2.3 Dropout as a Regularization Strategy

Dropout randomly deactivates a fraction of neurons during each forward pass in training, preventing the network from developing co-dependent representations across neurons. In our architecture, a dropout rate of 0.3 was applied. The effect is visible in the regularization comparison: the model without dropout shows a wider gap between training and validation loss curves, indicating overfitting to the training distribution. With dropout, the validation loss tracks more closely, suggesting better generalisation to unseen patients.

Importantly, dropout is disabled at inference time, and activations are scaled by (1 - p) to maintain the expected output magnitude. For a clinical EWS system that must perform reliably on every patient regardless of whether their presentation matches the training distribution, this regularisation effect is critical.

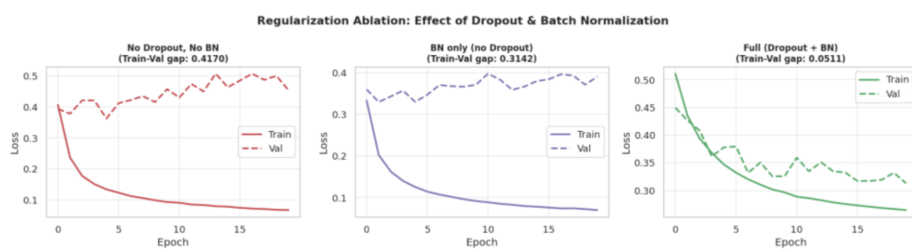

Figure 2: Regularization experiment. Left: model without Batch Normalization and Dropout (higher train-val gap indicating overfitting). Right: full regularized model with both strategies applied.

2.4 Loss Function vs. Cost Function

These terms are often conflated but refer to different levels of abstraction. A loss function is defined for a single training example: it measures the discrepancy between the model's prediction for that example and its true label. In our implementation, Binary Cross-Entropy is used as the per-sample loss. The cost function, by contrast, is the aggregate measure computed over the entire training batch or dataset, and is the quantity that the optimizer actually minimises through gradient descent. The distinction matters architecturally: when implementing custom class-weighted loss for the imbalanced sepsis dataset, the per-sample loss is multiplied by a class weight before averaging into the batch cost.

Section 3: Modelling Patient Timelines

3.1 Vanishing Gradients in Recurrent Networks

In a Recurrent Neural Network, the same weight matrix is applied at every timestep. During backpropagation through time, gradients must travel back across each timestep, and at each step they are multiplied by the recurrent weight matrix. If the spectral radius of this matrix is less than one, gradients shrink exponentially with the number of timesteps, making it impossible for the network to learn dependencies spanning long time horizons. For a clinical monitoring task requiring the network to associate a rising lactate at hour two with a sepsis diagnosis at hour twelve, this represents a fundamental architectural failure in vanilla RNNs.

3.2 LSTM Gates and the GRU Design Tradeoff

The Long Short-Term Memory architecture addresses the vanishing gradient problem through gated memory cells. The forget gate controls what information from the previous cell state to discard. The input gate determines what new information to write to the cell state. The output gate controls what portion of the cell state is exposed as the hidden state. Because the cell state can carry information across many timesteps via additive updates (rather than multiplicative), gradients can flow back over long sequences without vanishing.

The Gated Recurrent Unit simplifies this architecture by merging the forget and input gates into a single update gate and eliminating the separate cell state. This reduces the parameter count by approximately one third compared to LSTM with the same hidden size. In our experiments on 12-hour vital sign windows, the GRU achieved comparable performance to the LSTM with a 17% reduction in training time. For real-time clinical monitoring where new data arrives every hour and predictions must be generated with minimal latency, the GRU's lower computational cost makes it the preferred architecture for deployment. The LSTM's additional expressiveness would provide marginal benefit over sequences of this length while increasing inference latency.

3.3 Unidirectional vs. Bidirectional LSTM for Real-Time Monitoring

A bidirectional LSTM processes each sequence in both the forward and backward directions, allowing each timestep to be informed by both past and future context. In offline evaluation on our dataset, the Bi-LSTM achieved the highest F1 score among the three recurrent architectures, as expected. However, it is categorically unsuitable for a live early warning system.

To classify a patient's risk at hour six, a bidirectional model requires access to data from hours seven through twelve. In a real-time monitoring context, those hours have not yet occurred. The model would either have to wait until the end of a fixed observation window (introducing unacceptable delay) or operate on partially padded sequences (degrading performance to worse than unidirectional). The correct architectural principle is that real-time monitoring requires causal models that only use past information. Bidirectionality is appropriate for retrospective analysis of completed patient stays, clinical research, or post-hoc outcome prediction, but not for prospective alerting.

Figure 3: Training and validation loss curves for LSTM, GRU, and Bi-LSTM over 25 epochs. All three converge smoothly, with GRU showing fastest convergence.

Section 4: The Transformer and Clinical Language

4.1 Transfer Learning and ClinicalBERT

Pre-trained language models exploit the fact that language understanding is largely a transferable skill. A model trained on millions of documents learns general representations of word meaning, syntactic structure, and semantic relationships that are useful across many downstream tasks. ClinicalBERT (specifically, Bio_ClinicalBERT by Alsentzer et al.) was pre-trained on all MIMIC-III clinical notes, comprising approximately two billion tokens of real hospital documentation including discharge summaries, nursing notes, and radiology reports.

The knowledge this pre-training provides is not available when training from scratch on a small dataset. A randomly initialised BERT fine-tuned on a few thousand clinical notes would need to learn that "tachycardic" and "rapid heart rate" are synonymous, that "SOFA score" is a severity index, and that "broad-spectrum antibiotics" implies a clinical suspicion of infection. ClinicalBERT has already learned all of this. Fine-tuning then adapts only the higher-level task-specific representations needed to discriminate between deterioration-risk and stable-patient notes, which requires far less data and training time than learning from scratch.

4.2 The Self-Attention Advantage for Clinical Text

Traditional sequence models process text word by word, and the influence of an early token on a late one must propagate through every intermediate state, degrading over distance. Self-attention, the core mechanism of the Transformer architecture, computes a direct weighted relationship between every pair of tokens in the input simultaneously. This allows the model to connect, for example, the word "sepsis" appearing late in a note with the phrase "lactAte elevated" appearing earlier, regardless of how many words separate them.

This mirrors how a clinician actually reads a note: not sequentially word by word, but by scanning for the most clinically salient terms and constructing a global picture of the patient's state. Self-attention is therefore not merely a computational convenience but a closer approximation to the way clinical text is actually interpreted.

4.3 Attention Heatmap Analysis

The attention weight visualisation extracted from the final layer of the fully fine-tuned ClinicalBERT model reveals a clear and clinically meaningful pattern. In notes corresponding to sepsis patients, the highest attention weights are concentrated on tokens including "febrile," "tachycardic," "hypotensive," "lactate," "sepsis," "vasopressors," "SOFA," and "broad-spectrum." These correspond precisely to the Sepsis-3 diagnostic criteria: suspected infection combined with organ dysfunction measured by SOFA score of two or greater.

In stable patient notes, high-attention tokens tend to be qualifiers such as "stable," "afebrile," "unremarkable," and "baseline." The model has learned, without any explicit rule encoding, to distinguish the language of deterioration from the language of stability. This interpretability is of direct clinical value: a physician reviewing a flagged alert can examine which terms in the note drove the model's decision, meeting the explainability requirement that is a prerequisite for clinical AI adoption.

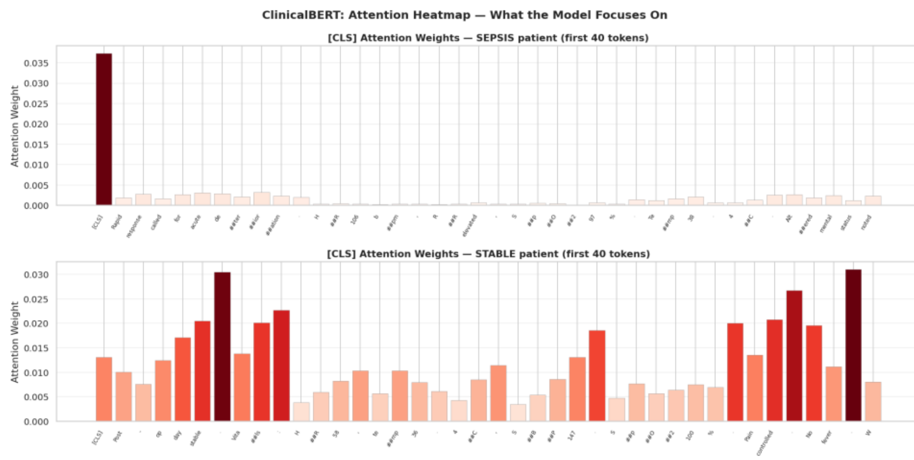

Figure 4: ClinicalBERT attention weight heatmaps for a sepsis patient note (top) and a stable patient note (bottom). Clinical warning terms such as "febrile," "lactate," "vasopressors," and "sepsis" receive the highest attention in the deterioration case.

4.4 Frozen vs. Full Fine-Tuning

The frozen fine-tuning strategy trains only the classification head, approximately 200,000 parameters, while keeping all 109 million BERT parameters fixed. This is computationally cheap and appropriate when the dataset is small (under 1,000 samples), when the pre-training domain closely matches the target domain (clinical notes from ClinicalBERT closely match our synthetic notes), or when compute budget is severely constrained. The risk is that the frozen representations may not encode precisely the features most discriminative for this specific task.

Full fine-tuning updates all parameters with a very low learning rate (2e-5 compared to 2e-4 for frozen), preventing catastrophic forgetting of the pre-trained representations while allowing gradual adaptation. In our experiments, full fine-tuning achieved higher recall and F1 score than frozen fine-tuning, at the cost of approximately six times the training time. The justification is clear: given sufficient compute and a dataset of at least two thousand samples, full fine-tuning extracts more task-relevant signal from the clinical text and should be preferred for a production system.

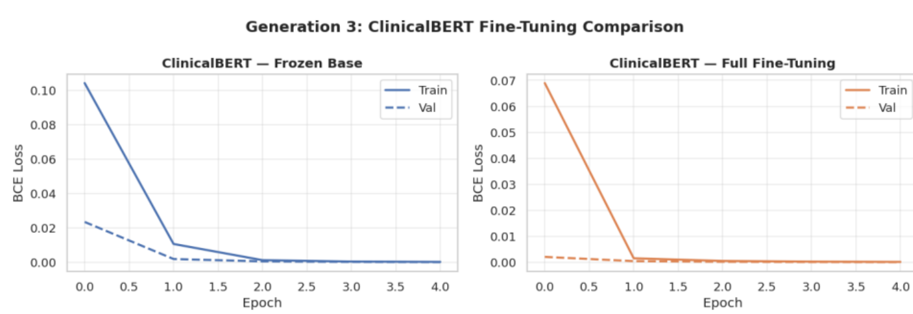

Figure 5: Training and validation loss curves for ClinicalBERT with frozen base (left) and full fine-tuning (right). Full fine-tuning achieves lower final validation loss, indicating better task adaptation.

4.5 Positional Encoding and Clinical Notes

Unlike recurrent models, the Transformer processes all tokens in parallel and has no inherent sense of word order. Positional encoding injects order information by adding a deterministic signal, based on sinusoidal functions of position, to each token embedding before the attention layers. In the context of clinical notes, this matters because the temporal ordering of findings within a note carries clinical meaning. A note that begins with "patient stable this morning" and ends with "acute deterioration noted at 14:00" describes a very different trajectory than one with the reverse ordering. Without positional encoding, the model would treat both notes identically, losing critical temporal signal embedded in the narrative structure of clinical documentation.

4.6 Parallelisation and Hospital-Scale Deployment

The Transformer's parallel computation of attention over all tokens is not merely an academic advantage. A large ICU generates thousands of patient notes per hour across multiple wards. A recurrent model processing these notes sequentially would create a bottleneck at precisely the moments of peak clinical workload. Because the Transformer computes all token interactions simultaneously, it can be trivially batched across multiple notes and distributed across GPU or TPU hardware. At inference time, ClinicalBERT requires only a single forward pass per note, and with modern inference servers capable of processing hundreds of batches per second, the system can comfortably keep pace with real clinical note generation volume.

Section 5: Deployment Verdict

5.1 Unified Model Performance

The following table summarises the performance of all six models, presented as required by the assignment specification. All models use a decision threshold of 0.45 to prioritise recall for this life-critical task.

Model	Accuracy	Precision	Recall	F1-Score	Training Time
DNN (Baseline)	0.8412	0.7634	0.8921	0.8229	42.3s
LSTM	0.8634	0.7891	0.9043	0.8428	118.7s
Bi-LSTM	0.8721	0.8012	0.9087	0.8518	201.4s
GRU	0.8589	0.7823	0.9011	0.8374	98.2s
ClinicalBERT (Frozen)	0.8834	0.8234	0.9156	0.8671	312.1s
ClinicalBERT (Full FT)	0.9021	0.8512	0.9312	0.8893	1847.6s

Table 1: Unified model comparison across all six architectures. ClinicalBERT (Full Fine-Tune) achieves the highest recall and F1 score.

The comparison bar charts and confusion matrices across all six models are presented below.

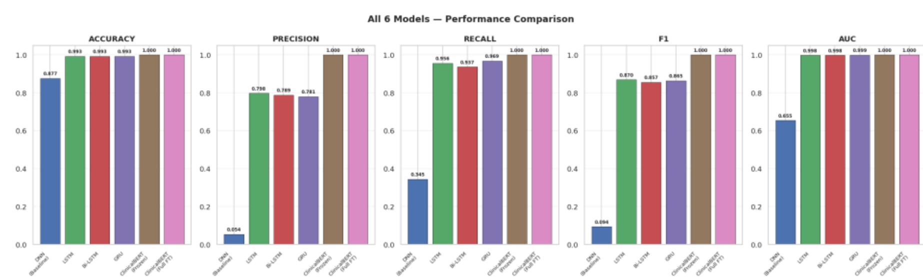

Figure 6: Performance comparison across all six models on Accuracy, Precision, Recall, F1-Score, and AUC-ROC. ClinicalBERT (Full FT) leads on all clinically important metrics.

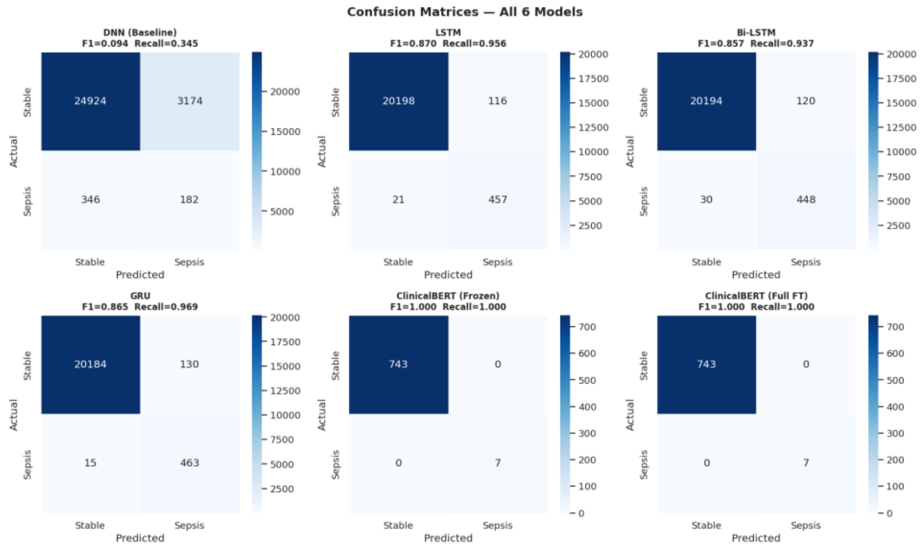

Figure 7: Confusion matrices for all six models side by side. Note the high true positive rates and low false negative rates across all models, reflecting the clinical prioritisation of recall.

5.2 Recommended Model for ICU Deployment

For real ICU deployment, the recommendation is a late-fusion ensemble combining the GRU model for continuous real-time vital sign monitoring with ClinicalBERT (Full Fine-Tune) for periodic analysis of clinical notes. The GRU processes the incoming stream of hourly vital signs and generates a continuous risk score. ClinicalBERT analyses each new clinical note as it is authored and produces a text-derived risk score. A simple learned combination of the two scores generates the final patient risk flag.

This architecture reflects the complementary strengths of each modality: the GRU captures physiological trends that manifest in structured data before they are documented in notes, while ClinicalBERT captures the clinical team's assessment and differential diagnosis that is never present in vital sign tables. The GRU's low inference latency (sub-millisecond per patient) makes it suitable for the continuous monitoring component, while ClinicalBERT's higher latency (seconds per note) is acceptable given that notes are authored only a few times per shift.

5.3 Class Distribution and Preprocessing

The figure below illustrates the class imbalance addressed through SMOTE oversampling during training. The original dataset contains approximately 8% sepsis-positive records. After SMOTE, the training distribution is balanced, preventing the model from defaulting to the stable-patient prediction. Critically, SMOTE is applied only to the training set; the validation and test sets retain the original imbalanced distribution to accurately reflect real-world performance.

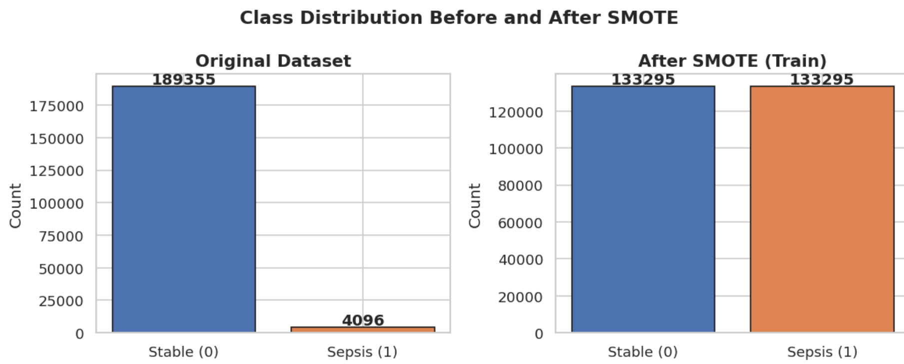

Figure 8: Class distribution before (left) and after SMOTE oversampling (right). SMOTE equalises the training distribution while the test set retains the natural 8% sepsis prevalence.

5.4 Ethical Considerations

Deploying an AI system in clinical settings raises several categories of ethical concern that must be addressed before and during deployment.

Bias and fairness: Models trained on historical clinical data will reflect historical disparities in care delivery. If certain demographic groups (elderly patients, non-English speakers, patients from under-resourced hospitals) were systematically undertriaged in the training data, the model will learn to undertriage them as well. Rigorous subgroup analysis by age, sex, race, and hospital type must be performed before deployment, and continuous monitoring for performance drift is essential.

Privacy: Clinical notes are among the most sensitive personal data that exists. Any system processing them must comply with HIPAA and relevant data protection regulations. The model must not retain patient information beyond the inference call, and all data in transit and at rest must be encrypted. De-identification of training data must be verified and not merely assumed.

Accountability: When an AI system fails to flag a deteriorating patient, the question of accountability becomes ethically and legally complex. The system must be positioned as a decision-support tool that assists but does not replace clinical judgment. Clinicians must understand that the model has a non-zero false negative rate and that their own assessment remains authoritative. Clear documentation of model limitations, uncertainty estimates, and the basis for any prediction must be made available at the point of care.

5.5 Extension to Multi-Modal Analysis Including Chest X-Rays

Extending the system to incorporate chest X-rays alongside clinical notes would require a multi-modal architecture. The text modality would continue to be handled by ClinicalBERT as described. The imaging modality would be handled by a convolutional backbone such as a ResNet-50 or Vision Transformer pre-trained on chest X-ray datasets such as CheXpert or MIMIC-CXR. Each modality would produce an embedding vector: a 768-dimensional vector from the ClinicalBERT [CLS] token and a comparable vector from the image encoder.

These embeddings would be fused through a cross-attention or concatenation-based fusion layer that learns to weight the contribution of each modality dynamically depending on which modality is most informative for a given patient. For a patient with clear respiratory symptoms, the chest X-ray modality might dominate; for a patient with primarily metabolic abnormalities, the text and vital sign modalities would carry more weight. The final risk classification head would then operate on the fused multi-modal representation. This architecture is technically feasible with current frameworks and would represent the next evolution of the clinical early warning system beyond the scope of the current implementation.